Sarah Sandmann[1], Yvonne Lisa Behrens[2], Gudrun Göhring[2] and Julian Varghese[1]

[1]Institute of Medical Informatics, University of Münster, Germany

[2]Department of Human Genetics, Hannover Medical School, Germany

Institut für Medizinische Informatik

Plaice plots – an allele-aware visualization of clonal evolution

Background

Reconstruction of clonal evolution involves complex integrated analyses. The results are, in addition to classical representation by phylogenetic or clonal evolution trees, commonly visualized using fish plots (e.g. R packages 'fishplot' [1] and 'timescape' [2]). In these plots, the development of every individual clone is displayed, considering time on the x-axis, and cancer cell fraction on the y-axis. Thereby, fish-shaped objects are generated.

Despite providing a comprehensive visualization of clonal evolution, fish plots display information only on clone-, not on allele-level. Biallelic mutations cannot be identified at first sight. However, with respect to disease progression, these mutations play an essential role [3].

Methods and Results

To improve visualization of biallelic mutations, we introduce plaice plots as a derivative of fish plots. The actual 'fish' become flatfish, i.e. plaice, and are mirrored – above and below the y-axis. The upper plot visualizes common clonal development, while the lower plot shows the fraction of remaining healthy alleles. For example, in case of mutated *TP53* and additional del17 affecting the remaining healthy allele, the fraction of cells with deficient *TP53* is marked in the lower plot. Similarly, X-chromosomal mutations in male samples, leading to a loss of the only available healthy allele, are visualized.

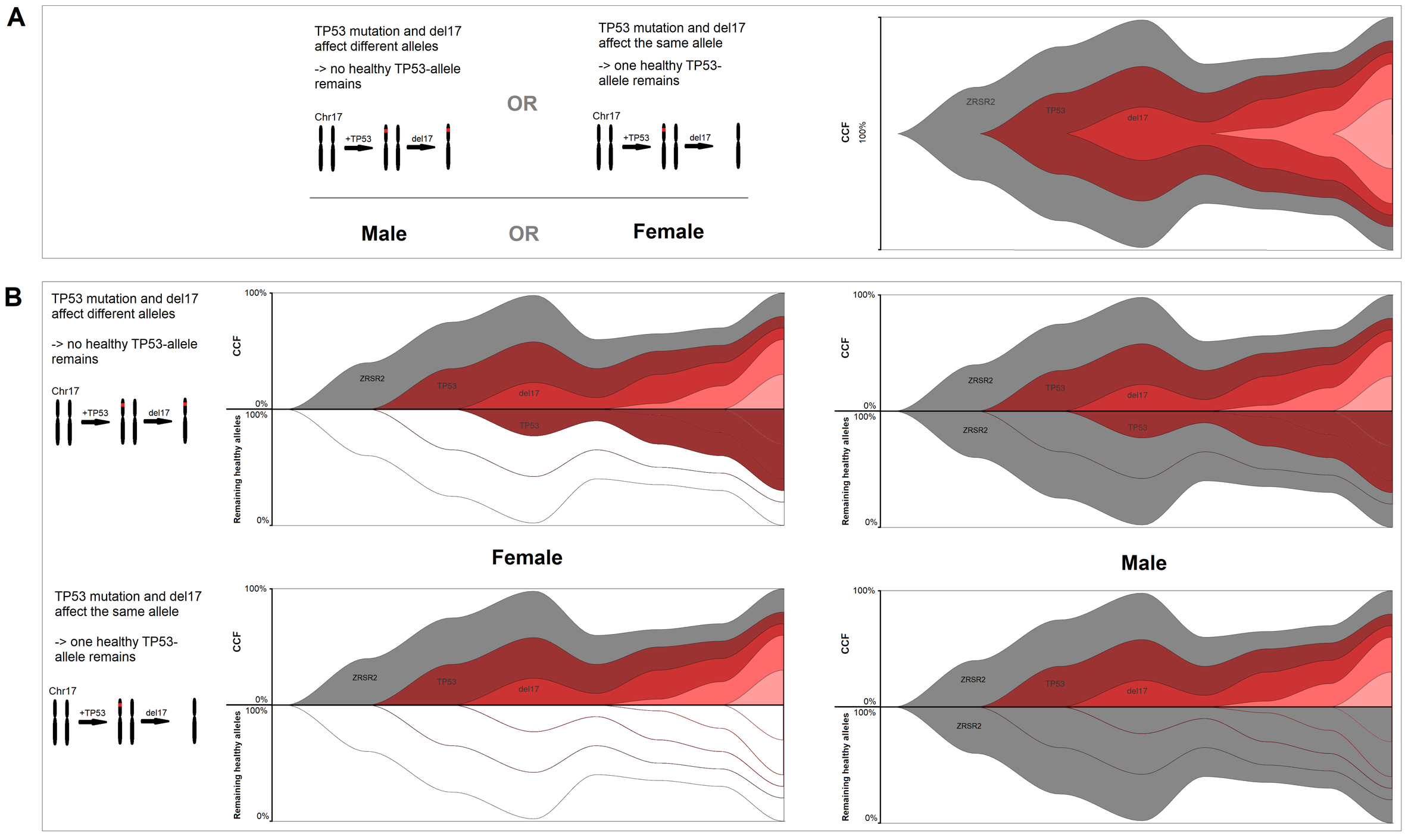

Figure 1: Exemplary visualization of clonal evolution using A) common fish plots, B) plaice plots. A *TP53*-mutation and del17 may affect the same or different alleles. Furthermore, a mutation in *ZRSR2*, located on the X-chromosome, has a different effect dependent on whether the studied subject is male or female. For all four scenarios, the same fish plot is generated. By contrast, differences can be observed using plaice plots.

Discussion

Common fish plots provide a comprehensive visualization of clonal evolution, allowing for analysis of clonal development over time and identification of dominant clones. However, information on allele-level is lacking. By contrast, plaice plots contain all information on clonal development just like fish plots and, in addition, information on remaining healthy alleles. Thereby, plaice plots allow for immediate identification of double-hit events.

References

[1] Miller, C.A. *et al.* (2016). Visualizing tumor evolution with the fishplot package for R. *BMC Genomics*, **17**(880). **[2]** Smith, M. *et al.* (2022). timescape: Patient Clonal Timescapes. *R package version 1.20.0*. **[3]** Reutter, K. *et al.* (2021). Reconstructing clonal evolution in relapsed and non-relapsed Burkitt lymphoma. *Leukemia*, **35**(2).

Contact: Sarah.Sandmann@uni-muenster.de

UKM Universitätsklinikum Münster

Westfälische Wilhelms-Universität Münster

medizinische fakultät Westfälische Wilhelms-Universität Münster